

MAT 426, Class 14: Householder QR Factorization

Two facts about QR Factorization using Gram-Schmidt Orthogonalization:

- The number of FLOPS required to find the QR Factorization using Gram-Schmidt orthogonalization is _____.
- The algorithm is _____ and can result in possible _____ loss of _____.

Definition. A matrix of the form

$$H =$$

where $\mathbf{u}$ is a _____ vector in $\mathbb{R}^n$ is called a **Householder matrix**.

Example 1. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

(a) Find the associated Householder matrix H .

(b) For any $\mathbf{x} \in \mathbb{R}^2$, compute $H\mathbf{x}$.

(c) For $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, sketch the vectors $\mathbf{u}$, $\mathbf{u}^\perp$, $\mathbf{x}$, and $H\mathbf{x}$ on the same set of axes.

PROPERTIES OF A HOUSEHOLDER MATRIX Let $H = I - \frac{2\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}}$ be a Householder matrix with $\mathbf{u} \in \mathbb{R}^n$. Then,

(a) H is symmetric; that is $H^T = H$

(b) H is orthogonal

(c) $\|H\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$

(d) $H^2 = I$

(e) $H\mathbf{u} = -\mathbf{u}$

(f) $H\mathbf{v} = \mathbf{v}$ if $\mathbf{v}^T\mathbf{u} = 0$

(g) If $\mathbf{u}$ is parallel to a vector $\mathbf{x} - \mathbf{y}$, where $\mathbf{y} \neq \mathbf{x}$ but $\|\mathbf{y}\| = \|\mathbf{x}\|$, then $H\mathbf{x} = \mathbf{y}$.

Proof.

When computing matrix-vector and matrix-matrix products involving a Householder matrix H , the matrix H does not have to be explicitly formed since

- $H\mathbf{x} =$
- $HA^T =$
- $AH =$

Here $\mathbf{x} \in \mathbb{R}^n$, A is an $m \times n$ matrix, and H an $n \times n$ Householder matrix. Also, $\beta = \frac{2}{\mathbf{u}^T\mathbf{u}}$.

THEOREM Given a nonzero vector $\mathbf{x} \neq \mathbf{e}_1$, the Householder matrix H defined by the vector $\mathbf{u} = \mathbf{x} \pm \|\mathbf{x}\|\mathbf{e}_1$ is such that $H\mathbf{x} = \mp\|\mathbf{x}\|\mathbf{e}_1$.

Proof.

Two Computational Considerations:

- The $\pm$ is chosen by _____ to avoid catastrophic cancellation in computing the first component of $\mathbf{u}$.
- To avoid any possibility of underflow or overflow in the computation of $\|\mathbf{x}\|$ the vector $\mathbf{u}$ can be determined by the vector _____ instead of $\mathbf{x}$ itself. (Our goal is to use this to find the QR decomposition and so we can always rescale our orthonormal vectors in Q without impacting orthogonality.)

Example 2. Let $\mathbf{x} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$. Create a Householder matrix H so that $H\mathbf{x}$ is a multiple of $\mathbf{e}_1$.

QR Factorization via Householder Matrices

Idea: Reduce the matrix A to an _____ matrix R by successively _____ multiplying A with a series of _____ matrices.

Outline of this process for a 4×3 matrix A

Example 3. Find the QR factorization of $A = \begin{bmatrix} 1 & 1 \\ 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}$ using the Householder QR algorithm.